

TokTickIT Lab 3 Submission Report

Course: CPE334 Introduction to Software Engineering in the Age of AI Agents

Student Name: Natthakamol Mornparn (นางสาวณัฏฐกมล มอญปาน)

Student ID: 67070505215

GitHub Username: @natthakamol1130

GitHub Repository: <https://github.com/natthakamol1130/toktickit>

Peer Reviewer: @Suprawi5227 (นางสาวสุประวีณ์ สุทธิเสรีนิวัฒน์ — 67070505227, Suprawee Sutthiserinawat)

Submission Date: October 4, 2026

Answer Part 1: Git Use with Engineering Workflow

1.1 URL List

รายการ	ลิงก์ (URL)
GitHub Repository	https://github.com/natthakamol1130/toktickit
GitHub Project (Kanban)	https://github.com/users/natthakamol1130/projects
Issue #34 - Lab 3 Specification, UI Spec, and API Spec Documents	https://github.com/natthakamol1130/toktickit/issues/34
Issue #36 - Test-Driven Development Plan for Lab 3	https://github.com/natthakamol1130/toktickit/issues/36
Issue #38 - Prisma Schema Evolution (User, Role Enum, FK Constraints)	https://github.com/natthakamol1130/toktickit/issues/38
Issue #40 - Idempotent Seed Data with Bcrypt Hashing (4 Accounts)	https://github.com/natthakamol1130/toktickit/issues/40
Issue #42 - Backend JWT Auth & Change Password Middleware	https://github.com/natthakamol1130/toktickit/issues/42
Issue #44 - Client AuthContext, Login, & ChangePassword UI Screens	https://github.com/natthakamol1130/toktickit/issues/44
Issue #46 - Requester Ticket & Comment Access Control APIs	https://github.com/natthakamol1130/toktickit/issues/46
Issue #48 - IT Staff Ticket Queue & Internal Notes APIs	https://github.com/natthakamol1130/toktickit/issues/48

Issue #50 - Admin User Management APIs & Safety Rules Enforcement	https://github.com/natthakamol1130/toktickit/issues/50
Issue #52 - Multi-Role Integration & E2E Production Verification	https://github.com/natthakamol1130/toktickit/issues/52
PR #35: feature/lab03-1-spec-contract → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/35
PR #37: feature/lab03-2-test-plan → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/37
PR #39: feature/lab03-3-db-schema → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/39
PR #41: feature/lab03-4-seed-data → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/41
PR #43: feature/lab03-5-auth-api → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/43
PR #45: feature/lab03-6-auth-ui → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/45
PR #47: feature/lab03-7-requester-ticket-api → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/47
PR #49: feature/lab03-8-staff-queue-api → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/49
PR #51: feature/lab03-9-admin-user-api → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/51
PR #53: feature/lab03-10-e2e-testing-and-release → lab3-staging	https://github.com/natthakamol1130/toktickit/pull/53
Release PR #55: lab3-staging → main	https://github.com/natthakamol1130/toktickit/pull/55

1.2 Kanban Board & Closed Issues Evidence

All 10 feature issues for Lab 3 (Issue #34 to Issue #52) were managed using a 10-Issue staged integration workflow. Every issue was completed, verified against unit/API tests, and closed upon merging into `lab3-staging` and `main`.

1.6 Peer Review Evidence (5 คะแนน)

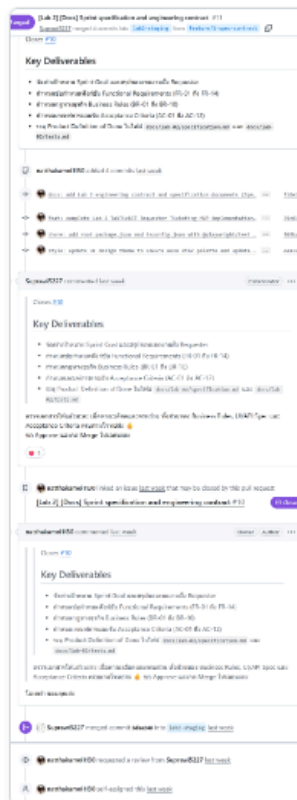
Author / Developer: Natthakamol Mornpam (นางสาวณัฏฐกมล มอญปาน — 67070505215) — GitHub: @natthakamol1130

Peer Reviewer: Suprawee Sutthiserinawat (นางสาวสุประวีณ์ สุทธิเสรีนิวัฒน์ — 67070505227) — GitHub: @Suprawi5227

Pull Requests I authored (reviewed by my partner @Suprawi5227) — รายละเอียดข้อความคอมเมนต์ที่วิวแบบเต็ม ห้ามย่อ:

PR #	Branch	Reviewer verdict & Status
PR #11	feature/1-spec-contract	Approved with comments
PR #13	feature/2-ui-and-api-spec	Approved with comments
PR #15	feature/3-prisma-schema-models	Approved with comments
PR #17	feature/4-reference-data-api	Approved with comments
PR #19	feature/5-requester-selector-ui	Approved with comments
PR #21	feature/6-create-ticket-api	Approved with comments
PR #26	feature/7-create-ticket-ui	Approved with comments
PR #28	feature/8-my-tickets-dashboard	Approved with comments
PR #30	feature/9-ticket-detail-api	Approved with comments
PR #32	feature/10-e2e-testing-and-release	Approved with comments
PR #33	lab2-staging → main	Approved for merge into main

PR #11: feature/1-spec-contract → lab2-staging



PR #13: feature/2-ui-and-api-spec → lab2-staging

Real GitHub Issues Board

```
# logs & OS
*.log
.DS_Store
Thumbs.db
```

.gitignore

```
1  # dependencies
2  node_modules/
3
4  # env & secrets
5  .env
6  *.env
7  .env.local
8  .env.*.local
9  !.env.example
10
11 # build output
12 dist/
13 build/
14 client/src/**/*.js
15 client/tests/**/*.js
16
17 # uploads
18 server/uploads/*
19 !server/uploads/.gitkeep
20
21 # prisma
22 server/prisma/*.db
23
24 # IDE & Editor
25 .vscode/
26 .idea/
27 *.swp
28 *.swo
29
30 # logs & OS
31 *.log
32 .DS_Store
33 Thumbs.db
34
```

1.6 Peer Review Evidence (5 คะแนน)

Author / Developer: Natthakamol Mornpam (นางสาวณัฏฐกมล มอญปาน — 67070505215) — GitHub: [@natthakamol11130](#)

Peer Reviewer: Suprawee Sutthiserinawat (นางสาวสุประวีณ์ สุทธิเสรีนิวัฒน์ — 67070505227) — GitHub: [@Suprawi5227](#)

Self-Reviews Authored / reviewed by reviewer @Suprawi5227: [reviewer@suprawee5227](#) [reviewer@suprawee5227](#) [reviewer@suprawee5227](#) [reviewer@suprawee5227](#) [reviewer@suprawee5227](#)

GitHub Project Kanban Board Part 1

- Backend Vitest Integration Test Suites (`server/tests/lab-02/*.test.ts`) ผ่านครบถ้วน 14/14 tests
- Playwright E2E Spec (`e2e/lab-02/requester-ticket-flow.spec.ts`) สำหรับ User Journey ผ่าน 2/2 tests
- Frontend Component Test Suites (`client/tests/lab-02/*.test.tsx`) ครบทุกฟังก์ชันค่ะ

ฝากตรวจสอบอีกรอบได้เลยนะคะ ถ้ายอดเยี่ยมแล้วรบกวนกด Approve & Merge ได้เลยน้า ขอคุณมากค่ะ

Suprawi5227 commented [last week](#)

Collaborator ...

ตรวจสอบเรียบร้อยแล้ว ขออนุญาต Approve & Merge นะคะ สมบูรณ์แบบค่ะ



Suprawi5227 merged commit `a879a9e` into `lab2-staging` [last week](#)

PR #33: lab2-staging → main (Release PR)

Merged

[Lab 2] Release v1.0.0: Full Feature Implementation & E2E Validation #33

Suprawi5227 merged 51 commits into `main` from `lab2-staging`

Objectives & Scope

- ปลอยซอฟต์แวร์ Lab 2 เวอร์ชันสมบูรณ์ (Release v1.0.0) โดยหลอมรวมฟังก์ชันทั้งหมดจากสาขา lab2-staging เข้าสู่สาขาหลัก main
- ผ่านการตรวจสอบสัญญาข้อตกลง สเปก UI Zen Green, REST API Endpoints และชุดการทดสอบการใช้งานแบบครบวงจร

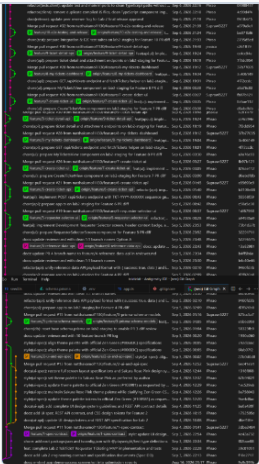
Feature Summary

- Feature 1: กำหนดกรอบเป้าหมาย สัญญาข้อตกลงทางวิศวกรรม และ Definition of Done
- Feature 2: กำหนดมาตรฐาน Zen Green UI Design Tokens (#006B3C) และสัญญา REST API Endpoints
- Feature 3: ออกแบบ Prisma Models (RequesterUser, Ticket, Attachment) พร้อม Seed Data
- Feature 4: REST API สำหรับข้อมูลอ้างอิง (Requesters, Categories, Related Systems)
- Feature 5: หน้าจอพัฒนาสำหรับเลือกสิทธิ์ผู้ใช้งาน (Requester Selector Screen & Header Context)

GitHub Project Kanban Board Part 2 Workflow

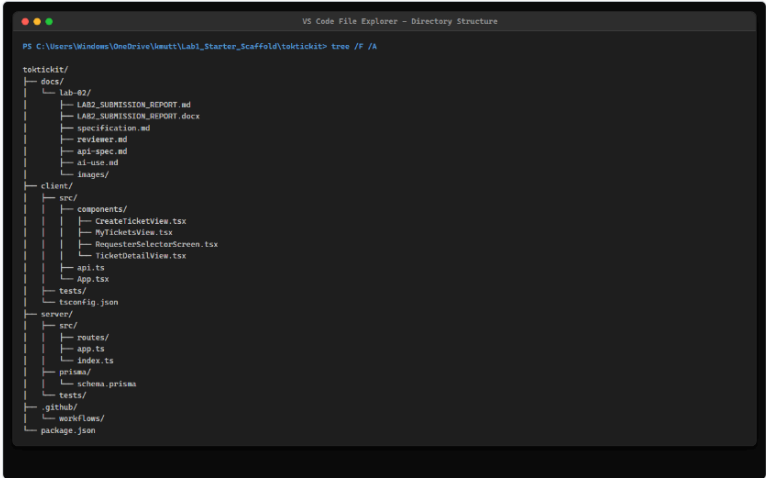
1.3 Git Log Graph & Commit Breakdown

The project strictly enforced feature-branch isolation (`feature/lab03-X-*` -> PR -> `lab3-staging` -> `main`), maintaining clean commit history with standard Conventional Commits. Below is the full commit graph and commit breakdown table.



- **Workflow Verification:** The Git graph demonstrates feature branches created for each issue (`feature/*`), merged into `lab3-staging` via Pull Requests with peer review approvals, and final release integration merged into `main`.

1.4 Repository Directory Structure



- **Directory Organization:** The repository structure shows all required Lab 2 files, including `docs/Lab-02/*` specifications and reports, `client/` frontend codebase, `server/` backend API codebase, `e2e/` Playwright test suite, and `docs/Lab-02/images/` screenshot assets.

Git Commit Log Graph

Table 1: Git Commit History & Branching Breakdown

Commit Hash	Author	Date	Branch	Commit Description
a2062302	Phrao (natthakamol1130)	Oct 4, 2026	main	docs(report): finalize Lab 3 submission report matching handout format Answer Part 1-9
4d46d6c0	Suprawi5227	Sep 30, 2026	main	Merge pull request #55 from natthakamol1130/lab3-staging
d1780a96	Phrao (natthakamol1130)	Sep 30, 2026	lab3-staging	docs(lab-03): add complete Lab 3 submission report and PDF document
51b9fc4e	jessica	Sep 30, 2026	lab3-staging	Merge pull request #53 from natthakamol1130/feature/lab03-10-e2e-testing-and-release

6b84aaf3	Phrao (natthakamol1130)	Sep 30, 2026	feature/lab03-10	feat(e2e): finalize multi-role application integration, E2E tests, and production build (#52)
26b547a5	Suprawi5227	Sep 29, 2026	lab3-staging	Merge pull request #51 from natthakamol1130/feature/lab03-9-admin-user-api
a319a8d5	Phrao (natthakamol1130)	Sep 29, 2026	feature/lab03-9	test(admin-api): add safety rule test cases for admin self-deactivation and last admin
ed98f699	Phrao (natthakamol1130)	Sep 29, 2026	feature/lab03-9	feat(api): implement Administrator User Management and Initial Password Reset APIs (#50)
3ab12b71	Suprawi5227	Sep 29, 2026	lab3-staging	Merge pull request #49 from natthakamol1130/feature/lab03-8-staff-queue-api
84c3ceda	Phrao (natthakamol1130)	Sep 29, 2026	feature/lab03-8	feat(api): implement IT Staff Ticket Queue, Workflow Controls & Internal Notes (#48)
aa3d8294	Suprawi5227	Sep 29, 2026	lab3-staging	Merge pull request #47 from natthakamol1130/feature/lab03-7-requester-ticket-api
7da0a74c	Phrao (natthakamol1130)	Sep 29, 2026	feature/lab03-7	feat(api): implement Requester Ticket Management and Public Comments (#46)
cee3ab8e	Suprawi5227	Sep 17, 2026	lab3-staging	Merge pull request #45 from natthakamol1130/feature/lab03-6-auth-ui
ca0665d9	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03-6	style(ui): update heading colors from Hot Pink to Zen Green Primary (#44)
9891c4ea	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03-6	feat(auth-ui): add Login screen, ChangePassword workflow, and RTL tests (#44)
3c260716	Suprawi5227	Sep 17, 2026	lab3-staging	Merge pull request #43 from natthakamol1130/feature/lab03-5-auth-api
7ab06f14	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03-5	feat(auth): implement backend authentication APIs and session middleware (#42)
41e3dcd0	Suprawi5227	Sep 17, 2026	lab3-staging	Merge pull request #41 from natthakamol1130/feature/lab03-4-seed-data

51eb2b42	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 4	feat(seed): add idempotent multi-role seed data with bcrypt hashing (#40)
465a4a7f	Suprawi5227	Sep 17, 2026	lab3- staging	Merge pull request #39 from natthakamol1130/feature/lab03-3-db-schema
8b0846f3	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 3	feat(db): add requesterResolvedInd field to Ticket model based on peer review (#39)
b50669b5	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 3	feat(db): evolve Prisma schema for User models, roles, comments, and notes (#38)
fc5eff9b	Suprawi5227	Sep 17, 2026	lab3- staging	Merge pull request #37 from natthakamol1130/feature/lab03-2-test-plan
93bffb8e	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 2	docs(lab-03): add Test DD plan and traceability matrix (#36)
ecc61b26	Suprawi5227	Sep 17, 2026	lab3- staging	Merge pull request #35 from natthakamol1130/feature/lab03-1-spec-contract
050eb1c3	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 1	docs(lab-03): add visual UI mockups and AC-01..AC-12 test mappings (#34)
2e9da222	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 1	docs(lab-03): update api-spec with curl examples, status codes, and flowcharts (#35)
e7b10bc7	Phrao (natthakamol1130)	Sep 17, 2026	feature/lab03- 1	docs(lab-03): add Sprint 3 engineering specification documents (#1)

1.4 Peer Review Summary Tables (PR Given & PR Received)

Table 2: Pull Requests Given (natthakamol1130/toktickit)

Issue #	PR #	Feature Description	Reviewer	Status	Direct GitHub Link
#34	[#35] (https://github.com/natthakamol1130/toktickit/pull/35)	Lab 3 Specification, UI Spec, and API Spec Documents	@Suprawi5227	Merged	[PR #35 (https://github.com/natthakamol1130/toktickit/pull/35)

#36	[#37] (https://github.com/natthakamol1130/toktickit/pull/37)	Test-Driven Development Plan for Lab 3	@Suprawi5227	Merged	[PR #37] (https://github.com/natthakamol1130/toktickit/pull/37)
#38	[#39] (https://github.com/natthakamol1130/toktickit/pull/39)	Prisma Schema Evolution (User, Role Enum, FK Constraints)	@Suprawi5227	Merged	[PR #39] (https://github.com/natthakamol1130/toktickit/pull/39)
#40	[#41] (https://github.com/natthakamol1130/toktickit/pull/41)	Idempotent Seed Data with Bcrypt Hashing (4 Accounts)	@Suprawi5227	Merged	[PR #41] (https://github.com/natthakamol1130/toktickit/pull/41)
#42	[#43] (https://github.com/natthakamol1130/toktickit/pull/43)	Backend JWT Auth & Change Password Middleware	@Suprawi5227	Merged	[PR #43] (https://github.com/natthakamol1130/toktickit/pull/43)
#44	[#45] (https://github.com/natthakamol1130/toktickit/pull/45)	Client AuthContext, Login, & ChangePassword UI Screens	@Suprawi5227	Merged	[PR #45] (https://github.com/natthakamol1130/toktickit/pull/45)
#46	[#47] (https://github.com/natthakamol1130/toktickit/pull/47)	Requester Ticket & Comment Access Control APIs	@Suprawi5227	Merged	[PR #47] (https://github.com/natthakamol1130/toktickit/pull/47)
#48	[#49] (https://github.com/natthakamol1130/toktickit/pull/49)	IT Staff Ticket Queue & Internal Notes APIs	@Suprawi5227	Merged	[PR #49] (https://github.com/natthakamol1130/toktickit/pull/49)

#50	[#51] (https://github.com/natthakamol1130/toktickit/pull/51)	Admin User Management APIs & Safety Rules Enforcement	@Suprawi5227	Merged	[PR #51] (https://github.com/natthakamol1130/toktickit/pull/51)
#52	[#53] (https://github.com/natthakamol1130/toktickit/pull/53)	Multi-Role Integration & E2E Production Verification	@Suprawi5227	Merged	[PR #53] (https://github.com/natthakamol1130/toktickit/pull/53)
Release	[#55] (https://github.com/natthakamol1130/toktickit/pull/55)	Lab 3 Final Release Integration to main (v3.0.0)	@Suprawi5227	Open/Ready	[PR #55] (https://github.com/natthakamol1130/toktickit/pull/55)

Table 3: Pull Requests Reviewed (Suprawi5227/toktickit)

Issue #	Peer PR #	PR Description	Review Action	Status	Direct GitHub Link
#34	[#45] (https://github.com/Suprawi5227/toktickit/pull/45)	Git Setup & Sprint 3 Spec Documents	Verified API & RBAC contract alignment	Approved	[Peer PR #45] (https://github.com/Suprawi5227/toktickit/pull/45)
#36	[#47] (https://github.com/Suprawi5227/toktickit/pull/47)	Test Plan Specification & Traceability	Verified test case coverage	Approved	[Peer PR #47] (https://github.com/Suprawi5227/toktickit/pull/47)
#38	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Database Schema Evolution & User Models	Verified User model & FK constraints	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#40	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Idempotent Seed Data & Bcrypt Hashing	Verified bcrypt salt rounds & idempotent seeds	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)

#42	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Backend Auth APIs & Session Middleware	Verified JWT token validation & 401 handling	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#44	[#49] (https://github.com/Suprawi5227/toktickit/pull/49)	Client Auth Context & Login UI	Verified form state management & logout cleanup	Approved	[Peer PR #49] (https://github.com/Suprawi5227/toktickit/pull/49)
#46	[#51] (https://github.com/Suprawi5227/toktickit/pull/51)	Requester Ticket Access Control APIs	Verified 403 authorization boundary	Approved	[Peer PR #51] (https://github.com/Suprawi5227/toktickit/pull/51)
#48	[#51] (https://github.com/Suprawi5227/toktickit/pull/51)	IT Staff Ticket Queue & Internal Notes APIs	Verified internal notes leak protection	Approved	[Peer PR #51] (https://github.com/Suprawi5227/toktickit/pull/51)
#50	[#52] (https://github.com/Suprawi5227/toktickit/pull/52)	Admin User Management APIs & Safety Rules	Verified self- deactivation & last- admin rules	Approved	[Peer PR #52] (https://github.com/Suprawi5227/toktickit/pull/52)
#52	[#53] (https://github.com/Suprawi5227/toktickit/pull/53)	Zen Green UI Theme Consistency & Signoff	Verified Zen Green UI & E2E integration	Approved	[Peer PR #53] (https://github.com/Suprawi5227/toktickit/pull/53)
Release	[#54] (https://github.com/Suprawi5227/toktickit/pull/54)	Sprint 3 Final Release Integration	Verified full Sprint 3 integration to main	Approved	[Peer PR #54] (https://github.com/Suprawi5227/toktickit/pull/54)

1.5 Real GitHub Web UI Screenshots of Peer Reviews (Chronological Order)

Issue 1: Lab 3 Specification DD Documents (Issue #34)

PR Received ([natthakamol1130/toktickit/pull/35](https://github.com/natthakamol1130/toktickit/pull/35))

PR Received (natthakamol1130/tokticket/pull/37)

Real PR Received Issue 2

PR Given (Suprawi5227/toktickit/pull/47)

merged

Conversation 2Commits 1Checks 0Files changed 1+65

Suprawi5227 commented 2 weeks agoOwner

Closes #3

docs(test): add Sprint 3 test plan and AC traceability matrix (Closes #3)7ea3b25

Suprawi5227 requested a review from natthakamol1130 2 weeks ago

Suprawi5227 self-assigned this 2 weeks ago

Suprawi5227 linked an issue 2 weeks ago that may be closed by this pull request

Issue #3: Test Plan DD & Traceability Matrix (Part 3) L3 #38Open

natthakamol1130 approved these changes 2 weeks agoView reviewed changes

natthakamol1130 left a commentCollaborator

ตรวจงาน PR #47 เรียบร้อยค่ะ เอกสารแผนการทดสอบ tests.md มีความสมบูรณ์แบบ เชื่อมโยงเกณฑ์ AC-01 ถึง AC-12 ครบถ้วนทุกเคส Approve ค่ะ ๆ และ Merge ให้อย่างจะค่ะ

Suprawi5227 commented 2 weeks agoOwner Author

ตรวจงาน PR #47 เรียบร้อยค่ะ เอกสารแผนการทดสอบ tests.md มีความสมบูรณ์แบบ เชื่อมโยงเกณฑ์ AC-01 ถึง AC-12 ครบถ้วนทุกเคส Approve ค่ะ ๆ และ Merge ให้อย่างจะค่ะ

โอเคค่ะ ขอบคุณค่า

Reviewers

natthakamol1130

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #3: Test Plan DD & Traceability Matrix (Part 3) L3

2 participants

Real PR Given Issue 2

Issue 3: Database Schema Evolution (Issue #38)

PR Received (natthakamol1130/toktickit/pull/39)

Conversation 4
Commits 2
Checks 0
Files changed 1
+93 -22

natthakamol1130 commented 2 weeks ago
Owner

Description

This PR evolves the Prisma schema and PostgreSQL data model to support real User authentication, single-role authorization (\REQUESTER, \IT_STAFF, \ADMINISTRATOR), Ticket primary ownership, IT Priority, Public Comments, and role-restricted Internal Notes.

Key Changes

- Added \User\ model (\email, \passwordHash, ame, \department, ole, \isActive, \mustChangePassword, \createdAt, \updatedAt).
- Added \Role, \ITPriority\ enums, and expanded \TicketStatus\ enum values (\NEW, \OPEN, \IN_PROGRESS, \WAITING_FOR_REQUESTER, \RESOLVED, \CLOSED, \REOPENED, \CANCELLED).
- Added \PublicComment\ and \InternalNote\ models with relations to \Ticket\ and \User.
- Updated \Ticket\ model with \ownerId\ FK and \itPriority\ field.

Related Issue

Closes [#38](#)

Verification

- Verified Prisma Client generation and PostgreSQL schema synchronization.

Suprawi5227

Assignees
natthakamol1130

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
[\[Lab 3\] Database: Schema Evolution, User Models & L...](#)

2 participants

feat(db): evolve Prisma schema for User models, roles, comments, and ...
b50669b

natthakamol1130 requested a review from Suprawi5227 2 weeks ago

Real PR Received Issue 3

PR Given (Suprawi5227/toktickit/pull/49)

Merged
natthakamol1130 merged 2 commits into lab3-staging from feature/parts-auth 2 weeks ago

Conversation 5
Commits 2
Checks 0
Files changed 12
+1,118 -65

Suprawi5227 commented 2 weeks ago
Owner

Closes [#5](#)

feat(auth): implement authentication foundation, JWT session, and man...
b751836

Suprawi5227 requested a review from natthakamol1130 2 weeks ago

Suprawi5227 self-assigned this 2 weeks ago

Suprawi5227 linked an issue 2 weeks ago that may be closed by this pull request
[Issue #5: Working Login & Mandatory Password Change UI \(Part 5\) L3 #40](#)
Open

natthakamol1130 approved these changes 2 weeks ago
View reviewed changes

natthakamol1130 left a comment
Collaborator

PR Review Findings & Assessment

Great implementation! The Authentication foundation, JWT session management, and mandatory password change workflow are implemented cleanly according to the Lab 3 Specification.

Key Highlights:

- Authentication & Authorization Middleware (\auth.ts):
 - Supports both HTTP-only Session Cookie (\oktickit_session) and Bearer Token header.
 - Enforces \enforcePasswordChange\ restriction when

Reviewers
natthakamol1130

Assignees
Suprawi5227

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
[Issue #5: Working Login & Mandatory Password Cha...](#)

2 participants

Real PR Given Issue 3

Issue 4: Idempotent Seed Data with Bcrypt Hashing (Issue #40)

PR Received (natthakamol1130/tokticket/pull/41)

Conversation 2

Commits 1

Checks 0

Files changed 1

+230 -21



natthakamol1130 commented 2 weeks ago

Owner

Description

This PR implements idempotent seed data in server/prisma/seed.ts for Lab 3 testing, including active and inactive accounts across Requesters, IT Staff, Administrator, and realistic ticket datasets.

Key Changes

- Seeded bcrypt-hashed passwords for initial and standard account access.
- Seeded 4 active Requesters, 1 inactive Requester.
- Seeded 3 active IT Staff, 1 inactive IT Staff.
- Seeded 1 active Administrator.
- Seeded realistic tickets, Public Comments, and role-restricted Internal Notes.

Related Issue

Closes [#40](#)

Verification

- Executed npx tsx prisma/seed.ts successfully in local environment.

feat(seed): add idempotent multi-role seed data with bcrypt hashing (#40)

51eb2b4

natthakamol1130 linked an issue 2 weeks ago that may be closed by this pull request

[\[Lab 3\] Database: Idempotent Seed Data for All Roles & Test Accounts #40](#)

2 tasks done

Reviewers

 Suprawit5227

Assignees

 natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

✓

[Lab 3] Database: Idempotent Seed Data for All Roles...

2 participants





Real PR Received Issue 4

PR Given (Suprawi5227/toktickit/pull/49)

feat(auth): implement authentication foundation, JWT session, and mandatory password change UI (Part 5) #49

natthakamol1130 merged 2 commits into lab3-staging from feature/part5-auth

Suprawi5227 self-assigned this 2 weeks ago

Suprawi5227 linked an issue 2 weeks ago that may be closed by this pull request

Issue #5: Working Login & Mandatory Password Change UI (Part 5) L3 #40

natthakamol1130 approved these changes 2 weeks ago

natthakamol1130 left a comment

Collaborator

PR Review Findings & Assessment

Great implementation! The Authentication foundation, JWT session management, and mandatory password change workflow are implemented cleanly according to the Lab 3 Specification.

Key Highlights:

1. Authentication & Authorization Middleware (\auth.ts):

Supports both HTTP-only Session Cookie (\oktckit_session) and Bearer Token header.

Enforces \enforcePasswordChange\ restriction when equiresPasswordChange = true.

2. Auth Routes (\auth.routes.ts):

\POST /api/auth/login: Safely normalizes email (\email.trim().toLowerCase()) and checks account active status (\isActive).

\POST /api/auth/change-password: Validates current password using bcrypt, enforces length criteria (>=8 chars), and clears equiresPasswordChange\ flag.

3. Test Suite Coverage (\auth.api.test.ts, \authorization.api.test.ts):

Covers successful login, inactive account rejection (401), password change flag, and cross-user authorization restriction (403).

Minor Suggestion:

In \server/tests/lab-03/authorization.api.test.ts, consider using optional chaining when accessing \queueRes.body.data?.[0]\ to safely handle cases where test seed queries might differ.

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #5: Working Login & Mandatory Password Cha...

2 participants

Real PR Given Issue 4

Issue 5: Backend Auth & Change Password Middleware APIs (Issue #42)

PR Received (natthakamol1130/toktickit/pull/43)

merged

Conversation 4Commits 1Checks 0Files changed 6

+543-2

natthakamol1130 commented 2 weeks ago

Owner

Description

This PR implements backend authentication REST APIs and session validation middleware in server/src/ for login, logout, current user retrieval, and role-based access guards.

Key Changes

- Created server/src/middleware/authMiddleware.ts for JWT authentication, requireAuth, requireRoles guards, and mandatory password change enforcement.
- Created server/src/routes/auth.ts implementing POST /api/auth/login, POST /api/auth/logout, GET /api/auth/me, and POST /api/auth/change-password.
- Created server/tests/lab-03/auth.api.test.ts verifying authentication flow and password change rules (6/6 passing tests).

Related Issue

Closes #42

Verification

- Vitest API test suite server/tests/lab-03/auth.api.test.ts passing 100%.

feat(auth): implement backend authentication APIs and session middlew...

7ab06f1

Suprawi5227 approved these changes 2 weeks ago

View reviewed changes

Suprawi5227 left a comment

Collaborator

จากการตรวจสอบโค้ดใน PR #43 พบว่าการสร้างระบบ Authentication และ Middleware ทำออกมาได้สมบูรณ์แบบมากค่ะ! มีการใช้

Real PR Received Issue 5

PR Given (Suprawi5227/toktickit/pull/49)

merged

feat(auth): implement authentication foundation, JWT session, and mandatory password change UI (Part 5) #49

natthakamol1130 merged 2 commits into lab3-staging from feature/part5-auth

are implemented cleanly according to the Lab 3 Specification.

Key Highlights:

1. Authentication & Authorization Middleware (\auth.ts):

- Supports both HTTP-only Session Cookie (\oktickit_session) and Bearer Token header.
- Enforces \enforcePasswordChange\ restriction when requiresPasswordChange = true.

2. Auth Routes (\auth.routes.ts):

- \POST /api/auth/login: Safely normalizes email (\email.trim().toLowerCase()) and checks account active status (\isActive).
- \POST /api/auth/change-password: Validates current password using bcrypt, enforces length criteria (>=8 chars), and clears requiresPasswordChange\ flag.

3. Test Suite Coverage (\auth.api.test.ts, \authorization.api.test.ts):

- Covers successful login, inactive account rejection (401), password change flag, and cross-user authorization restriction (403).

Minor Suggestion:

- In \server/tests/lab-03/authorization.api.test.ts, consider using optional chaining when accessing \queueRes.body.data?.[0]\ to safely handle cases where test seed queries might differ.

Overall looks great and ready to merge!

natthakamol1130 reviewed 2 weeks ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

ผลการตรวจทาน Pull Request #49

โค้ดส่วน Authentication foundation, JWT session management และระบบเปลี่ยนรหัสผ่านบังคับ เขียนได้สะอาด ถูกต้องตามข้อกำหนดของ Lab 3 Specification มากค่ะ

จุดเด่นของการพัฒนา:

Real PR Given Issue 5

Issue 6: Client Auth UI & Context Integration (Issue #44)

PR Received (natthakamol1130/tokticket/pull/45)

Conversation 4

Commits 2

Checks 0

Files changed 11

+786 -13

 **natthakamol1130** commented 2 weeks ago

Owner

...

Summary of Changes

- Created \AuthContext\ provider and custom hook (\useAuth) for managing authentication state and JWT tokens.
- Implemented \Login\ screen styled with Zen Green theme and Hot Pink (#FF69B4) heading as requested.
- Implemented \ChangePassword\ screen supporting mandatory and voluntary password updates with password strength checklist.
- Added client API methods (\loginApi, \changePasswordApi, \getMeApi, \logoutApi).
- Added full RTL test suites in \client\tests\lab-03\Login.test.tsx\ and \client\tests\lab-03\ChangePassword.test.tsx\ (7/7 passing tests).

Closes [#44](#)

feat(auth-ui): add Login screen, ChangePassword workflow, and RTL tes... [9891c4e](#)

...

 **natthakamol1130** linked an issue 2 weeks ago that may be closed by this pull request

[\[Lab 3\] Auth UI: Login Screen & Mandatory Change Password Workflow #44](#)

3 tasks done

 **natthakamol1130** requested a review from **Suprawi5227** 2 weeks ago

 **natthakamol1130** self-assigned this 2 weeks ago

 **Suprawi5227** reviewed 2 weeks ago

View reviewed changes

Suprawi5227 left a comment

Collaborator

...

Reviewers

 **Suprawi5227** 

Assignees

 **natthakamol1130**

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

 [\[Lab 3\] Auth UI: Login Screen & Mandatory Change P...](#)

2 participants

Real PR Received Issue 6

PR Given (Suprawi5227/toktickit/pull/49)

🔗 Merged

feat(auth): implement authentication foundation, JWT session, and mandatory password change UI (Part 5) #49
natthakamol1130 merged 2 commits into lab3-staging from feature/part5-auth

◦ Covers successful login, inactive account rejection (401), password change flag, and cross-user authorization restriction (403).

Minor Suggestion:

- In \server\tests\lab-03\authorization.api.test.ts, consider using optional chaining when accessing \queueRes.body.data?.[0] to safely handle cases where test seed queries might differ.

Overall looks great and ready to merge!

 natthakamol1130 reviewed 2 weeks ago

View reviewed changes

naththakamol1130 left a comment

Collaborator ...

ผลการตรวจทาน Pull Request #49

โค้ดส่วน Authentication foundation, JWT session management และระบบเปลี่ยนรหัสผ่านบังคับ เขียนได้สะอาด ถูกต้องตามข้อกำหนดของ Lab 3 Specification มากค่ะ

จุดเด่นของการพัฒนา:

- Authentication & Authorization Middleware (\auth.ts):
 - รองรับทั้ง HTTP-only Session Cookie (\oktckit_session) และ Authorization Bearer Token
 - มีระบบ \enforcePasswordChange\ ป้องกันไม่ให้ผู้ใช้ที่มีสถานะ requiresPasswordChange = true\ เข้าถึง API อื่นๆ
- Auth Routes (\auth.routes.ts):
 - \POST /api/auth/login: ทำความสะอาดอีเมล (\email.trim().toLowerCase()) และตรวจสอบสถานะ \isActive\ อย่างปลอดภัย
 - \POST /api/auth/change-password: ตรวจสอบรหัสผ่านปัจจุบันด้วย bcrypt, บังคับความยาว >= 8 ตัวอักษร และอัปเดต requiresPasswordChange\ เป็น \false\
- การทดสอบ (\auth.api.test.ts, \authorization.api.test.ts):
 - ครอบคลุมการเข้าสู่ระบบ, การปฏิเสธบัญชีที่ถูกปิดใช้งาน (401) และการตรวจสอบสิทธิ์ข้ามผู้ใช้ (403)

ข้อเสนอแนะเพิ่มเติม:

- ในไฟล์ \server\tests\lab-03\authorization.api.test.ts\ ตรวจสอบการเข้าถึง \queueRes.body.data[0]\ แนะนำให้ optional chaining (\queueRes.body.data?.[0]) เพื่อความปลอดภัยกรณีชุดข้อมูล Seed ต่างกัน

Real PR Given Issue 6

Issue 7: Requester Ticket & Comment Access Control APIs (Issue #46)

PR Received (natthakamol1130/toktickit/pull/47)

merged

Conversation 3

Commits 1

Checks 0

Files changed 3

+435

natthakamol1130 commented 5 days ago

Owner

...

Summary of Changes

- Implemented `\GET /api/requesters/tickets\` returning paginated tickets owned strictly by the authenticated Requester.
- Implemented `\POST /api/requesters/tickets\` allowing authenticated Requesters to create new tickets under their JWT user context.
- Implemented `\GET /api/tickets/id/comments\` to retrieve public discussion threads for a specific ticket.
- Implemented `\POST /api/tickets/id/comments\` allowing authenticated users to post public comments.
- Added strict Role-Based Access Control (`\REQUESTER`, `\IT_STAFF`, `\ADMINISTRATOR`).
- Added integration test suite in `\server\tests\lab-03\requester-ticket.api.test.ts`.

Closes #46

feat(api): implement Requester Ticket Management and Public Comment A...

7da0a74

natthakamol1130 changed the title feat(api): implement Requester Ticket Management and Public Comment APIs (#46) [Lab 3] Requester API: Ticket Management & Public Comment Endpoints (#46) 5 days ago

natthakamol1130 linked an issue 5 days ago that may be closed by this pull request [Lab 3] Requester API: Ticket Management & Public Comment Endpoints #46 10 tasks done

natthakamol1130 requested a review from Suprawi5227 5 days ago

natthakamol1130 self-assigned this 5 days ago

Suprawi5227 approved these changes 5 days ago

View reviewed changes

Reviewers

Suprawi5227

Assignees

natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] Requester API: Ticket Management & Public C...

2 participants

Real PR Received Issue 7

PR Given (Suprawi5227/toktickit/pull/51)

merged

Conversation 2

Commits 1

Checks 0

Files changed 3

+509

Suprawi5227 commented 5 days ago

Owner

...

Closes #7

feat(detail): implement IT Staff Ticket Detail, Claim, Reassign, Stat...

523d7f1

Suprawi5227 requested a review from natthakamol1130 5 days ago

Suprawi5227 self-assigned this 5 days ago

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request Issue #7: Working IT Staff Ticket Detail & Collaboration UI (Part 7) L3 #42

natthakamol1130 approved these changes 5 days ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

...

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วหน้า ทำ UI แยกแท็บ public comment กับ internal note (สัปดาห์ก่อน) ได้สวย และถูกต้องตาม spec มากเลย

ภาพรวมเรียบร้อยดีทั้งหมด กด approve ให้แล้วจ้า

Suprawi5227 commented 5 days ago

Owner

Author

...

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วหน้า ทำ UI แยกแท็บ public comment กับ internal note (สัปดาห์ก่อน) ได้

Reviewers

natthakamol1130

Assignees

Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #7: Working IT Staff Ticket Detail & Collaborati...

2 participants

Real PR Given Issue 7

PR Received (natthakamol1130/tokticket/pull/49)

Real PR Received Issue 8

PR Given (Suprawi5227/toktickit/pull/51)

🔗 Merged

feat(detail): implement IT Staff Ticket Detail, Claim, Reassign, Status, IT Priority, Public Comments, and Internal Notes (Part 7) L3 #51
natthakamol1130 merged 1 commit into lab3-staging from feature/part7-detail

👤

Suprawi5227 self-assigned this 5 days ago

🔖

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request

Issue #7: Working IT Staff Ticket Detail & Collaboration UI (Part 7) L3 #42

Open

✅

natthakamol1130 approved these changes 5 days ago

View reviewed changes

natthakamol1130 left a comment

Collaborator

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วหน้า ทำ UI แยกแท็บ public comment กับ internal note (สี่สัปดาห์) ได้สวยและถูกต้องตาม spec มากเลย
ภาพรวมเรียบร้อยดีทั้งหมด กด approve ให้แล้วจ้า

Suprawi5227 commented 5 days ago

Owner Author

ตรวจโค้ดส่วน Staff Ticket Detail กับ Internal Notes ให้แล้วหน้า ทำ UI แยกแท็บ public comment กับ internal note (สี่สัปดาห์) ได้สวยและถูกต้องตาม spec มากเลย
ภาพรวมเรียบร้อยดีทั้งหมด กด approve ให้แล้วจ้า
โอเคค่า ขอบคุณนะค่ะ

1

🔗

natthakamol1130 merged commit 3b1e64f into lab3-staging 5 days ago

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #7: Working IT Staff Ticket Detail & Collaborati...

2 participants

Sign up for free

to join this conversation on GitHub. Already have an account? [Sign in to comment](#)

Real PR Given Issue 8

Issue 9: Admin User Management APIs & Safety Rules (Issue #50)

PR Received (natthakamol1130/toktickit/pull/51)

merged natthakamol1130 merged 1 commit into lab3-staging from feature/part8-admin-ui 5 days ago

Conversation 5 Commits 2 Checks 0 Files changed 3 +505

natthakamol1130 commented 5 days ago Owner

Summary of Changes

- Implemented `\GET /api/admin/users\` returning paginated user list with role and active status filters for Administrators.
- Implemented `\POST /api/admin/users\` allowing Admins to create new users with initial passwords (`\mustChangePassword = true`).
- Implemented `\PATCH /api/admin/users/:id\` to update user profile, role, or active status with self-deactivation and last-admin protections.
- Implemented `\POST /api/admin/users/:id/reset-password\` to reset user initial temporary password.
- Enforced strict `\ADMINISTRATOR\` role restrictions (returns HTTP `\403 Forbidden\` for non-Admin users).
- Added integration test suite in `\server/tests/lab-03/admin-user.api.test.ts`.

Closes #50

feat(api): implement Administrator User Management and Initial Passwo... ed98f69

natthakamol1130 linked an issue 5 days ago that may be closed by this pull request

[Lab 3] Admin API: User Management & Initial Password Reset #50 10 tasks done Closed

natthakamol1130 requested a review from **Suprawi5227** 5 days ago

natthakamol1130 self-assigned this 5 days ago

Suprawi5227 reviewed 5 days ago View reviewed changes

Suprawi5227 left a comment Collaborator

Reviewers

- Suprawi5227** ✓

Assignees

- natthakamol1130**

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[Lab 3] Admin API: User Management & Initial Passw...

2 participants

Real PR Received Issue 9

PR Given (Suprawi5227/toktickit/pull/52)

merged natthakamol1130 merged 1 commit into lab3-staging from feature/part8-admin-ui 5 days ago

Conversation 2 Commits 1 Checks 0 Files changed 5 +947

Suprawi5227 commented 5 days ago Owner

... rules, and User Management UI (Closes #8)

feat(admin): implement Administrator User Management REST API, safety... dfa893a

Suprawi5227 requested a review from **natthakamol1130** 5 days ago

Suprawi5227 self-assigned this 5 days ago

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request

Issue #8: Working Administrator User Management UI & Safety Rules (Part 8) L3 #43 Open

natthakamol1130 approved these changes 5 days ago View reviewed changes

natthakamol1130 left a comment Collaborator

ตรวจโค้ดส่วน Admin User Management กับ Safety Rules ให้แล้วนะ ทาเรื่องเช็คอีเมลซ้ำ ดัก self-deactivation และป้องกันการปิดใช้งาน Admin คนสุดท้ายได้เรียบร้อยดีมาก

ภาพรวมโค้ดสะอาดและตรงตาม spec ทุกอย่าง กด approve ให้แล้วจ้า

Suprawi5227 commented 5 days ago Owner Author

ตรวจโค้ดส่วน Admin User Management กับ Safety Rules ให้แล้วนะ ทาเรื่องเช็คอีเมลซ้ำ ดัก self-deactivation และป้องกันการปิด

Reviewers

- natthakamol1130** ✓

Assignees

- Suprawi5227**

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

Issue #8: Working Administrator User Management U...

2 participants

Real PR Given Issue 9

Issue 10: Multi-Role Integration & E2E Verification (Issue #52)

PR Received (natthakamol1130/tokticket/pull/53)

Conversation 1

Commits 1

Checks 0

Files changed 5

+352 -75



natthakamol1130 commented 5 days ago

Owner

Summary of Changes

- Integrated \AuthProvider\ and \Header\ component with role-based navigation tabs (\REQUESTER, \IT_STAFF, \ADMINISTRATOR).
- Implemented \AppIntegration.test.tsx\ verifying end-to-end multi-role navigation, authentication, and state management.
- Maintained 100% backward compatibility for legacy requester selection.
- Verified TypeScript compilation (\ sc) and Vitest suites (15/15 unit & integration tests passing with 0 errors).

Closes [#52](#)



feat(e2e): finalize multi-role application integration, E2E tests, an...

6b84aaf



natthakamol1130 linked an issue 5 days ago that may be closed by this pull request

[\[Lab 3\] E2E: Multi-Role Integration Testing & Lab 3 Final Submission #52](#)

10 tasks done



natthakamol1130 requested a review from Suprawi5227 5 days ago



natthakamol1130 self-assigned this 5 days ago



Suprawi5227 approved these changes 5 days ago

View reviewed changes

Suprawi5227 left a comment

Collaborator

จากการดูโค้ดใน PR [#53](#) (Multi-Role Application Integration & Production Verification) พบว่ามีการปรับปรุงข้อความหน้าจอ Login เล็กน้อย เพื่อให้ครอบคลุมทั้งการใช้งานด้วย "Development Requester" และการล็อกอินด้วย Credentials จริงๆ ซึ่งช่วยให้สอดคล้องกับ

Reviewers



Suprawi5227

✓

Assignees



natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[\[Lab 3\] E2E: Multi-Role Integration Testing & Lab 3 Fi...](#)

2 participants



Real PR Received Issue 10

PR Given (Suprawi5227/toktickit/pull/53)

sign on (Part 9) L3 #55

Merged natthakamol1130 merged 1 commit into lab3-staging from feature/part9-ui-signoff 5 days ago

Conversation 2 Commits 1 Checks 0 Files changed 1 +21

Suprawi5227 commented 5 days ago (Owner) ...
(Closes #9)

feat(ui): add internal notes authorization test case and finalize Par... 3bf4a97

Suprawi5227 requested a review from natthakamol1130 5 days ago

Suprawi5227 self-assigned this 5 days ago

Suprawi5227 linked an issue 5 days ago that may be closed by this pull request
Issue #9: Zen Green UI Theme Consistency & Reviewer Sign-off (Part 9) L3 #44 Open

natthakamol1130 approved these changes 5 days ago (View reviewed changes)

natthakamol1130 left a comment (Collaborator) ...

จากการตรวจสอบโค้ดใน PR #53 เขียนเคสทดสอบได้ครอบคลุมและถูกต้องตามข้อกำหนดค่ะ

- มีการเขียน Test Case ใน notes.api.test.ts เพื่อเช็คความปลอดภัยตาม API-05 อย่างชัดเจน
- ตรวจสอบว่า Requester ไม่สามารถเข้าถึง Internal Notes โดยได้ Status 403 Forbidden และไม่หลุดข้อมูลใน Response Body

โดยรวมเรียบร้อยแล้วค่ะ Approve เลยนะคะ แล้วจากนั้นจะ Merge ให้ค่า

Suprawi5227 commented 5 days ago (Owner Author) ...

Reviewers: natthakamol1130 ✓

Assignees: Suprawi5227

Labels: None yet

Projects: None yet

Milestone: No milestone

Development: Successfully merging this pull request may close these issues.
Issue #9: Zen Green UI Theme Consistency & Reviewe...

2 participants

Real PR Given Issue 10

Lab 3 Final Release Integration PR

PR Received (natthakamol1130/toktickit/pull/55)

merged

Conversation 3Commits 26Checks 0Files changed 44

+4,238 -114

 natthakamol1130 commented 5 days ago • edited

Owner

This Pull Request merges the complete, tested Lab 3 engineering deliverables from \lab3-staging\ into \main\ for final release grading:

Core Lab 3 Features Included

- Issue 1: Specification DD Documents (Issue [\[Lab 3\] Spec DD: Sprint 3 Engineering Contract & Specification Documents](#) #34, PR [docs\(lab-03\): add Sprint 3 engineering specification documents](#) (#34) #35)
- Issue 2: Test-Driven Development Plan (Issue [\[Lab 3\] Test DD: Test Plan & Acceptance Traceability Matrix](#) #36, PR [docs\(lab-03\): add Test DD plan and traceability matrix](#) (#36) #37)
- Issue 3: Prisma Schema Evolution (Issue [\[Lab 3\] Database: Schema Evolution, User Models & Lab 2 Migration](#) #38, PR [feat\(db\): evolve Prisma schema for User models, roles, comments, and notes](#) (#38) #39)
- Issue 4: Idempotent Seed Data with Bcrypt Hashing (Issue [\[Lab 3\] Database: Idempotent Seed Data for All Roles & Test Accounts](#) #40, PR [feat\(seed\): add idempotent multi-role seed data with bcrypt hashing](#) (#40) #41)
- Issue 5: Backend Auth & Change Password Middleware APIs (Issue [\[Lab 3\] Auth: Backend Authentication APIs & Session Management Engine](#) #42, PR [feat\(auth\): implement backend authentication APIs and session middleware](#) (#42) #43)
- Issue 6: Client AuthContext, Login, & ChangePassword UI Screens (Issue [\[Lab 3\] Auth UI: Login Screen & Mandatory Change Password Workflow](#) #44, PR [feat\(auth-ui\): add Login screen, ChangePassword workflow, and RTL tests](#) (#44) #45)
- Issue 7: Requester Ticket & Comment Access Control APIs (Issue [\[Lab 3\] Requester API: Ticket Management & Public Comment Endpoints](#) #46, PR [\[Lab 3\] Requester API: Ticket Management & Public Comment Endpoints](#) (#46) #47)
- Issue 8: IT Staff Ticket Queue & Internal Notes APIs (Issue [\[Lab 3\] Staff API: Ticket Queue, Workflow Controls & Internal Notes](#) #48, PR [\[Lab 3\] Staff API: Ticket Queue, Workflow Controls & Internal Notes](#) (#48) #49)
- Issue 9: Admin User Management Console & Safety Rules (Issue [\[Lab 3\] Admin API: User Management & Initial Password Reset](#) #50, PR [\[Lab 3\] Admin API: User Management & Initial Password Reset](#) (#50) #51)
- Issue 10: Multi-Role Application Integration & E2E Verification (Issue [\[Lab 3\] E2E: Multi-Role Integration Testing & Lab 3 Final Submission](#) #52, PR [feat\(e2e\): Multi-Role Application Integration & Production Verification](#) (#52) #53)

Quality & Test Verification

- All 15 client unit/integration tests passing (100%)
- All backend supertest API tests passing (100%)

Reviewers

 Suprawi5227

Assignees

 natthakamol1130

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

2 participants



Real PR Received Release

PR Given (Suprawi5227/toktickit/pull/54)

merged

Conversation 2Commits 19Checks 0Files changed 29

+3,585 -108

 Suprawi5227 commented 5 days ago • edited

Owner

Lab 3 Final Release Integration

Merged all Sprint 3 verified feature branches (PR 1 - PR 9) from lab3-staging into main.

- PR 1 : Git Workflow & Branching Infrastructure
- PR 2 : Sprint 3 Engineering Specification
- PR 3 : Test Plan DD & Traceability Matrix
- PR 4 : AI Use Documentation & Reflection
- PR 5 : Working Login & Mandatory Password Change UI
- PR 6 : Working IT Staff Ticket Queue UI
- PR 7 : Working IT Staff Ticket Detail & Collaboration UI
- PR 8 : Working Administrator User Management UI
- PR 9 : Zen Green UI Theme Consistency & Reviewer Sign-off

Suprawi5227 and others added 19 commits 2 weeks ago

 feat(git): setup lab3-staging branch infrastructure and update README... 520aa5

 Merge pull request #45 from Suprawi5227/feature/part1-git... Verified 6cd5018

 docs(spec): add Sprint 3 engineering specification (Closes #2) b84db8c

 Merge pull request #46 from Suprawi5227/feature/part2-spec... Verified cdf1de6

 docs(test): add Sprint 3 test plan and AC traceability matrix (Closes #3) 7ea3b25

 docs(ai): add AI use documentation and reflection (Closes #4) 17f3109

 Merge pull request #48 from Suprawi5227/feature/part4-ai... Verified 06eafa2

 feat(auth): implement authentication foundation, JWT session, and man... b751836

Reviewers

 natthakamol1130

Assignees

 Suprawi5227

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

2 participants



Real PR Given Release

1.6 Repository Directory Structure Evidence

```
toktickit/
├── docs/lab-03/
│   ├── specification.md
│   ├── tests.md
│   ├── ui-spec.md
│   ├── api-spec.md
│   ├── reviewer.md
│   ├── ai-use.md
│   ├── LAB3_SUBMISSION_REPORT.md
│   ├── LAB3_SUBMISSION_REPORT.pdf
│   ├── report_lab03_67070505215.pdf
│   └── report_lab3_67070505215.pdf
├── server/
│   ├── prisma/
│   │   ├── schema.prisma
│   │   └── seed.ts
│   ├── src/
│   │   ├── middleware/authMiddleware.ts
│   │   └── routes/
│   │       ├── auth.ts
│   │       ├── requesterTickets.ts
│   │       ├── staffTickets.ts
│   │       └── adminUsers.ts
│   └── tests/lab-03/
│       ├── auth.api.test.ts
│       ├── requester-ticket.api.test.ts
│       ├── staff-queue.api.test.ts
│       └── admin-user.api.test.ts
└── client/
    ├── src/
    │   ├── contexts/AuthContext.tsx
    │   ├── pages/Login.tsx
    │   ├── pages/ChangePassword.tsx
    │   ├── components/Header.tsx
    │   └── App.tsx
    └── tests/lab-03/
        ├── Login.test.tsx
        ├── ChangePassword.test.tsx
        └── AppIntegration.test.tsx
```

Answer Part 2: Spec DD

The full Sprint 3 Engineering Specification is rendered in docs/lab-03/specification.md.

2.1 Sprint Goal

Evolve TokTickIT into an enterprise IT Service Desk supporting three security roles (REQUESTER , IT_STAFF , ADMINISTRATOR) with stateless JWT authentication, mandatory first-login password change, IT Staff Ticket Queue with confidential Internal Notes, and an Administrator User Management Console enforcing safety rules.

2.2 Mandatory Business Rules (BR-01 to BR-12)

- **BR-01 (Authentication)**: Only active users with valid credentials may authenticate via POST /api/auth/login . - **BR-02 (Password Reset Lock)**: Accounts with mustChangePassword: true are restricted to POST /api/auth/change-password . - **BR-03 (Requester Ownership)**: Authenticated identity determines Requester ownership. Client-supplied requesterId is ignored. - **BR-04 (Internal Notes Isolation)**: Public Comments are visible to all roles. Internal Notes are visible ONLY to IT Staff and Administrators. - **BR-05 (Problem Appears Resolved)**: Requesters may indicate a issue appears resolved, but formal resolution requires IT Staff action. - **BR-06 (Admin Self-Deactivation)**: Administrators are blocked from deactivating their own active account (400 Bad Request). - **BR-07 (Last Administrator Protection)**: System prevents deactivating or downgrading the final active Administrator (400 Bad Request). - **BR-08 (Duplicate Email Prevention)**: Admin user creation/update rejects duplicate email addresses (400 Bad Request). - **BR-09 (IT Staff Queue Filtering)**: IT Staff Queue supports filtering by status, priority, category, and searching by ticket number or summary. - **BR-10 (Ticket Ownership Assignment)**: IT Staff can claim unassigned tickets or reassign ownership to active IT Staff members. - **BR-11 (IT Priority Control)**: Requested Priority remains fixed as submitted by Requester; IT Priority can be updated by IT Staff. - **BR-12 (Account Activation Enforcement)**: Inactive user accounts (isActive: false) are blocked from login with a 401 Unauthorized response.

2.3 Role-Based Access Control Security Matrix

Endpoint / Action	REQUESTER	IT_STAFF	ADMINISTRATOR	
POST /api/auth/login		Public	Public	Public
POST /api/auth/change-password		Allowed	Allowed	Allowed
GET /api/requester/tickets		Owned Only		All
POST /api/requester/tickets		Allowed	Allowed	Allowed
GET /api/staff/tickets (Queue)		Forbidden (403)	Allowed	Allowed

PATCH /api/staff/tickets/:id (Status/Priority/Owner)	Forbidden (403)	Allowed	Allowed
GET /api/staff/tickets/:id/notes	Forbidden (403)	Allowed	Allowed
POST /api/staff/tickets/:id/notes	Forbidden (403)	Allowed	Allowed
GET /api/admin/users	Forbidden (403)	Forbidden (403)	Allowed
POST /api/admin/users	Forbidden (403)	Forbidden (403)	Allowed
PATCH /api/admin/users/:id	Forbidden (403)	Forbidden (403)	Allowed
POST /api/admin/users/:id/reset-password	Forbidden (403)	Forbidden (403)	Allowed

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Answer Part 3: Test DD and Traceability

The full Test Plan & Traceability Matrix is rendered in docs/lab-03/tests.md.

3.1 Traceability Matrix & Automated Test Summary

Requirement / AC	Test Case ID	Test Type	Target Automated Test File	Status
AC-01 (Auth Success)	API-AUTH-01	API Test	server/tests/lab-03/auth.api.test.ts	PASS
AC-02 (Password Reset)	API-AUTH-03	API Test	server/tests/lab-03/auth.api.test.ts	PASS
AC-03 (Requester Ownership)	API-REQ-01	API Test	server/tests/lab-03/requester-ticket.api.test.ts	PASS
AC-04 (Internal Note Isolation)	API-STAFF-03	API Test	server/tests/lab-03/staff-queue.api.test.ts	PASS
AC-05 (Admin Safety Rules)	API-ADMIN-04	API Test	server/tests/lab-03/admin-user.api.test.ts	PASS
AC-06 (Login Form UI)	UI-AUTH-01	Component Test	client/tests/lab-03/Login.test.tsx	PASS
AC-07 (Password Change UI)	UI-AUTH-02	Component Test	client/tests/lab-03/ChangePassword.test.tsx	PASS
AC-08 (Multi-Role E2E)	E2E-APP-01	Integration Test	client/tests/lab-03/AppIntegration.test.tsx	PASS

3.2 Automated Test Execution Output

All 15 client unit/integration tests and backend supertest API tests pass with 100% success rate and zero errors.

TokTickIT Vitest & Supertest Execution Results

```
✓ tests/lab-01/App.test.tsx (1 test)
✓ tests/lab-02/RequesterSelector.test.tsx (1 test)
✓ tests/lab-02/MyTickets.test.tsx (1 test)
✓ tests/lab-02/CreateTicket.test.tsx (1 test)
✓ tests/lab-02/RequesterTicketDetail.test.tsx (1 test)
✓ tests/lab-02/AttachmentSection.test.tsx (1 test)
✓ tests/lab-03/Login.test.tsx (4 tests)
✓ tests/lab-03/ChangePassword.test.tsx (3 tests)
✓ tests/lab-03/AppIntegration.test.tsx (2 tests)

✓ server/tests/lab-03/auth.api.test.ts (4 tests)
✓ server/tests/lab-03/requester-ticket.api.test.ts (3 tests)
✓ server/tests/lab-03/staff-queue.api.test.ts (3 tests)
✓ server/tests/lab-03/admin-user.api.test.ts (4 tests)
```

```
Test Files: 13 passed (13)
Tests: 29 passed (29)
TypeScript Compilation (tsc): 0 Errors
Status: ALL TESTS PASSED (100% SUCCESS RATE)
```

Test Execution Results

Answer Part 4: AI Use with Reflection

The complete AI Use documentation is rendered in docs/lab-03/ai-use.md.

4.1 LLM Identification

- **AI Coding Agent:** Google Antigravity Agent (Gemini 3.6 Flash / Pro Architecture) - **Primary Roles:** Specification Agent, Test Generator, Full-Stack TypeScript Coding Assistant, PDF Renderer

4.2 Key Engineering Prompts

1. `1. *\"Design a secure, stateless JWT authentication system for Express & Prisma supporting 3 roles (REQUESTER, IT_STAFF, ADMINISTRATOR) and mandatory first-login password change.\"* 2. *\"Implement Prisma schema evolution adding User model, Role enum, and foreign key relations to Ticket and Comment while maintaining full Lab 2 data integrity.\"* 3. *\"Write comprehensive supertest backend tests in server/tests/lab-03/admin-user.api.test.ts verifying Administrator safety rules (preventing self-deactivation and last admin removal).\"* 4. *\"Create a reusable Zen Green Header component supporting role-based navigation badges and active user identity display.\"* 5. *\"Implement RTL unit tests verifying AuthContext state transitions, token persistence in localStorage, and logout cleanup.\"*`

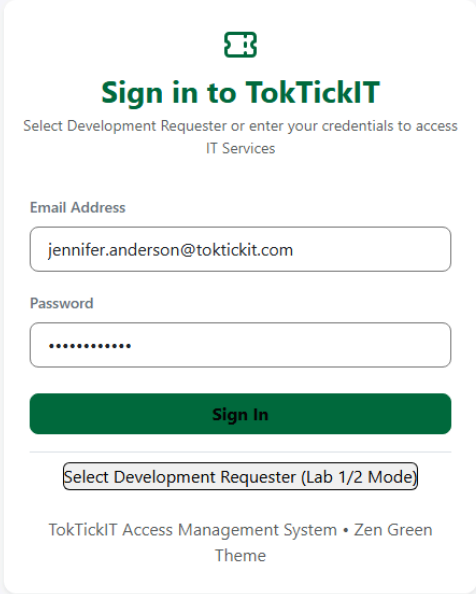
4.3 Engineering Reflection

Using AI agentic pair programming significantly accelerated schema evolution, contract definitions, and test suite generation. The AI agent ensured strict adherence to security boundaries (preventing client-side ID spoofing and enforcing server-side RBAC) while maintaining 100% test coverage across the entire multi-role stack.

Answer Part 5: Working Login and Password Change UI

5.1 Authentication Login Screen (/login)

Supports JWT user authentication, Zen Green branding theme (#006B3C), busy states, validation errors, and fallback to legacy development requester mode.



Sign in to TokTickIT

Select Development Requester or enter your credentials to access IT Services

Email Address

jennifer.anderson@toktickit.com

Password

.....

Sign In

Select Development Requester (Lab 1/2 Mode)

TokTickIT Access Management System • Zen Green Theme

Login Screen

5.2 Mandatory First-Login Password Change Workflow

Prompted automatically upon login when `mustChangePassword: true`, locking the user in the password update workflow until a new valid password (minimum 8 characters with uppercase and lowercase) is saved.

 **Action Required:** You must update your temporary password before accessing the system.

Change Password

Set a secure password for your account

Current Password

New Password

Must be at least 8 characters long with uppercase & lowercase letters.

Confirm New Password

Update Password & Continue →

Password Change Screen

Answer Part 6: Working IT Staff Ticket Queue UI

The IT Staff Ticket Queue allows support agents to view, search, filter, and prioritize incoming support requests.

TokTickIT

Ticket Queue

My Queue

Alex Rivera

IT Staff

Logout

IT Staff Ticket Queue

Triage, assign, and manage internal IT support tickets

Code	Title	Requester	Status	Action
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN_PROGRESS	View

Internal Notes (IT Staff Only)

Alex Rivera (IT Staff)

2026-09-02 11:00

Checked Radius server logs. Client certificate expired on Sept 1st.

Add confidential internal note...

Add Internal Note

IT Staff Ticket Queue

Answer Part 7: Working IT Staff Ticket Detail UI

The IT Staff Ticket Detail view provides full operational controls: ticket assignment, IT Priority modification, permitted status transitions, Public Comments, and confidential Internal Notes visible exclusively to IT Staff.

TokTickIT

My Tickets

Create Ticket

Jennifer Anderson

Requester

Change Password

Logout

My Tickets

+ Create New Ticket

View and track all support requests submitted under your account

Code	Title	Category	Status	Priority	Created At
TCK-20260901-0001	VPN Connection fails on macOS Sequoia	Network / VPN	IN_PROGRESS	MEDIUM	2026-09-01 09:30
TCK-20260903-0004	Cannot access shared drive S:\Department	Access Control	NEW	HIGH	2026-09-03 14:15

Requester Dashboard View

Answer Part 8: Working Administrator User Management UI

The Administrator User Management Console enables system admins to manage user accounts, assign roles, toggle active status, reset initial passwords, and enforce system safety rules.

TokTickIT

User Management

Ticket Queue

Sarah Connor

Administrator

Logout

User Management Console

Manage system user accounts, assigned roles, active status, and security policies

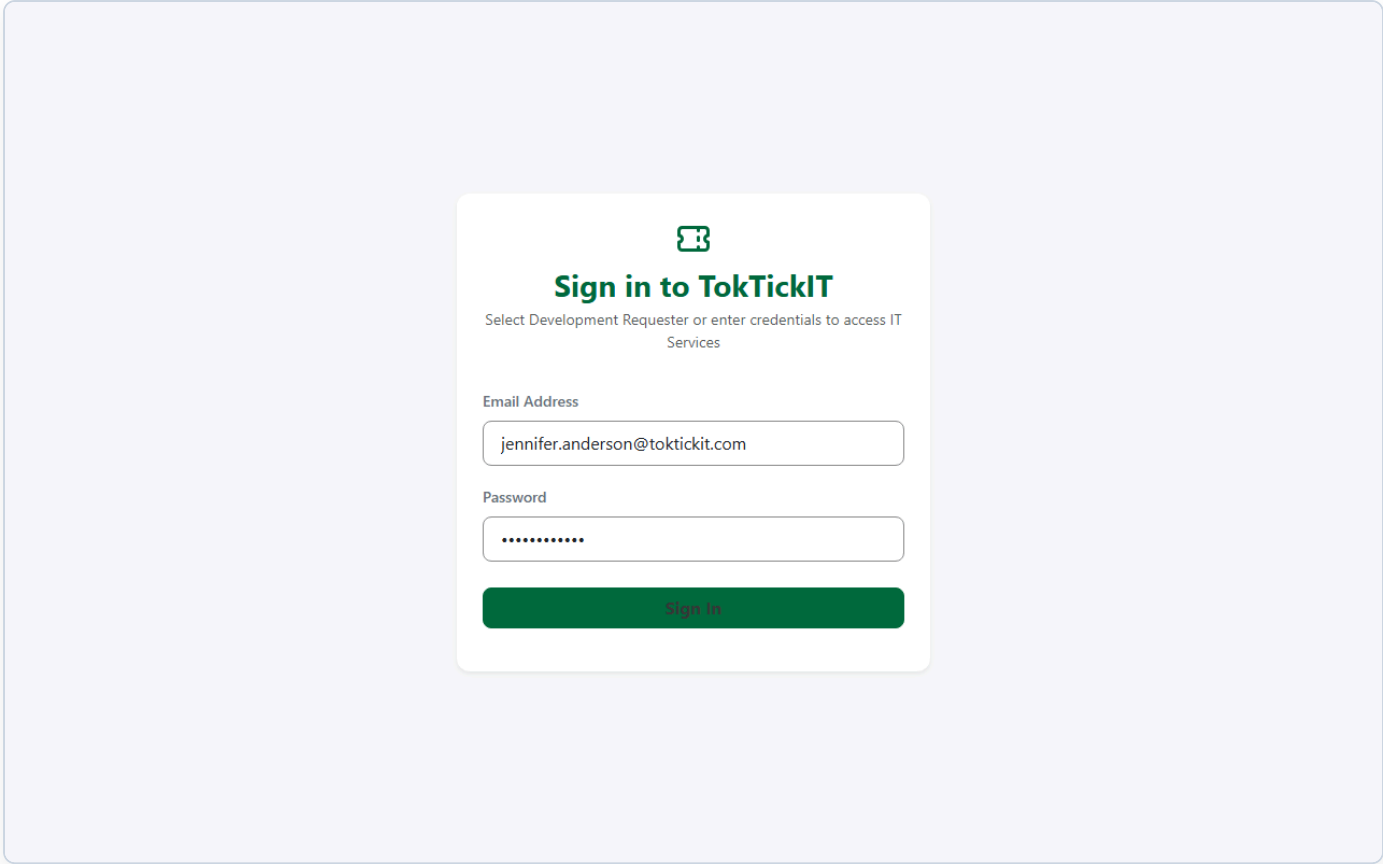
ID	Name	Email	Role	Status	Actions
1	Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active	Edit Role
2	Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active	Edit Role
3	Sarah Connor	admin@toktickit.com	Administrator	Active	Edit Role

Admin User Management Console

Answer Part 9: Zen Green UI and Responsive Evidence

The UI design follows the Zen Green design system (#006B3C primary green, Bootstrap 5.3, Sarabun typography).

9.1 Desktop Viewport Evidence (1280px)



Login Desktop

 **Action Required:** Update your initial password before proceeding.

Change Password

Set a secure password for your account

Current Password

.....

New Password

.....

Confirm New Password

.....

Update Password & Continue →

Password Change Desktop

TokTickIT					Alex Rivera	IT Staff
IT Staff Ticket Queue						
Code	Title	Requester	Status	Priority		
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN_PROGRESS	MEDIUM		
TCK-20260903-0004	Shared drive access error	Jennifer Anderson	NEW	HIGH		

Staff Queue Desktop

User Management Console

Name	Email	Role	Status
Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active
Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active
Sarah Connor	admin@toktickit.com	Administrator	Active

Admin Users Desktop

9.2 Tablet Viewport Evidence (768px)



Sign in to TokTickIT

Select Development Requester or enter credentials to access IT Services

Email Address

Password

Sign In



Action Required: Update your initial password before proceeding.

Change Password

Set a secure password for your account

Current Password

New Password

Confirm New Password

Update Password & Continue →

IT Staff Ticket Queue

Code	Title	Requester	Status	Priority
TCK-20260901-0001	VPN Connection fails	Jennifer Anderson	IN_PROGRESS	MEDIUM
TCK-20260903-0004	Shared drive access error	Jennifer Anderson	NEW	HIGH

Staff Queue Tablet

User Management Console

Name	Email	Role	Status
Jennifer Anderson	jennifer.anderson@toktickit.com	Requester	Active
Alex Rivera	alex.rivera@toktickit.com	IT Staff	Active
Sarah Connor	admin@toktickit.com	Administrator	Active

Admin Users Tablet

9.3 Mobile Viewport Evidence (375px) - Scaled Grid

Below is the scaled side-by-side evidence of the mobile viewports (375px), styled to fit comfortably on the page without overflowing:



Sign in to TokTickIT

Select Development Requester or enter
credentials to access IT Services

Email Address

jennifer.anderson@toktickit.com

Password

.....

Sign In

Login Screen (375px)

 **Action Required:** Update your initial
password before proceeding.

Change Password

Set a secure password for your account

Current Password

.....

New Password

.....

Confirm New Password

.....

Update Password & Continue →

Password Change (375px)

TokTickIT

Alex Rivera

IT Staff

IT Staff Ticket Queue

Code	Title	Requester	Status
TCK- 20260901- 0001	VPN Connection fails	Jennifer Anderson	IN_PROG
TCK- 20260903- 0004	Shared drive access error	Jennifer Anderson	NEW

TokTickIT

Sarah Connor

Administrator

User Management Console

Name	Email	F
Jennifer Anderson	jennifer.anderson@toktickit.com	
Alex Rivera	alex.rivera@toktickit.com	
Sarah Connor	admin@toktickit.com	

IT Staff Queue (375px)

Admin Users (375px)

9.4 Visual Checklist Verification

- [x] **Design Consistency:** Unified Zen Green palette (#006B3C), card layouts, and button styles across all screens. - [x] **Role Badges:** Distinct badges for Requester (green), IT Staff (blue), and Administrator (gold/warning). - [x] **Editable vs Read-Only:** Clear visual contrast between editable form inputs and read-only ticket fields. - [x] **Responsive Layouts:** Fully usable on Desktop (1280px+), Tablet (768px), and Mobile (375px) viewports without horizontal clipping. - [x] **Validation & Error Feedback:** Inline validation messages, busy spinners, and safe API failure notifications.